

Indices - Dividing Indices (Sheet 1)

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 Rewrite in expanded form.

Example $5^4 = 5 \times 5 \times 5 \times 5$

1 $7^5 = \square \times \square \times \square \times \square \times \square$

2 $4^6 = \square$

3 $3^4 = \square$

4 $2^3 = \square$

Now algebraic indices - same process

5 $k^3 = \square \times \square \times \square$

6 $b^4 = \square$

7 $m^5 = \square$

8 $c^2 = \square$

Use expansion and cancellation to answer these in index form.

Example

$$5^5 \div 5^2 = \frac{\cancel{5} \times \cancel{5} \times 5 \times 5 \times 5}{\cancel{5} \times \cancel{5}}$$

Cancel bottom with top

$$= 5 \times 5 \times 5 = 5^3$$

9 $3^6 \div 3^3 = \square$

$$= \square = \square$$

10 $\frac{y^6}{y^2} = \square$

$$= \square = \square$$

11 $a^3 \div a = \square$

$$= \square = \square$$

 Use expansion and cancellation to answer these, write a '1' on the top line when fully cancelled

Example

$$5^2 \div 5^5 = \frac{\cancel{5} \times \cancel{5}^1}{\cancel{5} \times \cancel{5} \times 5 \times 5 \times 5}$$

Cancel bottom with top

$$= \frac{1}{5 \times 5 \times 5} = \frac{1}{5^3}$$

12 $7^3 \div 7^5 = \square$

$$= \frac{\square}{\square} = \frac{\square}{\square}$$

13 $k \div k^5 = \square$

$$= \frac{\square}{\square} = \frac{\square}{\square}$$

14 $\frac{n^2}{n^4} = \square$

$$= \frac{\square}{\square} = \frac{\square}{\square}$$

You can divide indices by subtracting the indexes. Divide these, bigger power - smaller

Example

$$5^7 \div 5^4 = 5^{7-4} = 5^3$$

15 $3^5 \div 3^2 = \square$

16 $4^9 \div 4^4 = \square$

No power - it's like there is a '1' there

17 $8^6 \div 8 = \square$

18 $\frac{3^5}{3} = \square$

19 $6^8 \div 6 = \square$

Now algebraic indices - same process

20 $x^3 \div x = \square$

21 $\frac{n^7}{n^3} = \square$

22 $b^8 \div b^6 = \square$

23 $\frac{h^9}{h^4} = \square$

These will require a '1' up top, as the denominator is larger. Bigger power - smaller power.

$\frac{1}{\square}$

Example

$$\frac{h^4}{h^9} = \frac{1}{h^{9-4}} = \frac{1}{h^5}$$

24 $\frac{a^3}{a^7} = \square$

25 $5^5 \div 5^7 = \frac{5^5}{5^7} = \square$

26 $\frac{n}{n^6} = \square$

27 $\frac{e^3}{e^6} = \square$

28 $b^2 \div b^9 = \frac{b^2}{b^9} = \square$

29 $\frac{q^4}{q^5} = \square$

30 $c \div c^4 = \frac{c}{c^4} = \square$

31 $\frac{y^6}{y^8} = \square$